Amazon Electronics Market Intelligence Dashboard — Project Report

Data Visualization & Analytics (DVA) Capstone

Sector	Retail - Amazon Electronics
Team ID	E_G1
Team Members	Aman (Lead), Adnan Rizvi, Bhoomi Chhikara, Mouli Srivastava, Gauri Mishra, Prashant Raj, Swagato Bauri
GitHub Repository	E_G1_DVACapstone2
GitHub URL	https://github.com/Aman739-code/E_G1_DVACapstone2.git
Dataset	Amazon Product Sales Data (Kaggle)
Dataset Size	42,000 rows × 16 columns
Institute	Rishihood University
Submission Date	26th April 2026

1. Executive Summary

Problem

This project investigates which Amazon electronics product categories deliver the highest customer value through the combination of discount offered, rating received, and review engagement — and what pricing strategy drives the best performance.

Approach

The team sourced 42,000 Amazon product listings (16 attributes) from Kaggle and built a four-dashboard Tableau suite covering category overview, engagement & competitive analysis, pricing & discount dynamics, and customer value segmentation. Custom KPIs — Discount Effectiveness Score and High Value Product Rate — were engineered to answer the central decision question.

Key Insights

- High discounts do not reliably produce higher ratings; the relationship is weak across most categories.
- Only a small fraction of products simultaneously achieve high ratings, strong review volumes, and meaningful discounts — the true 'sweet spot'.
- Gaming, Storage & Memory, and Cameras & Photography show the highest average prices and most polarised discount patterns.
- Smartphones carry the largest average discount (22%+), yet Batteries and Office Supplies have among the lowest prices and discounts.
- Best-seller products account for only 0.73% of listings but show markedly different price profiles from the rest.

Key Recommendations

- Prioritise rating-first pricing strategies — customers respond to quality signals over discount depth.
- Focus promotional spend on the high-value sweet-spot segment identified by the Customer Value Scatter Plot.
- Category managers for Gaming and Storage & Memory should monitor the price-to-rating ratio as a primary health metric.

2. Sector and Business Context

Sector Overview

The Retail sector — specifically online electronics marketplaces — is one of the most data-rich and competitively dynamic environments for analytics. Amazon lists millions of electronics products, and margin management through discounting and visibility strategies is central to seller success.

Decision-Maker / Stakeholder

The primary stakeholders are Amazon sellers and category managers responsible for pricing, promotional spend allocation, and product portfolio decisions across 24+ electronics sub-categories.

Why This Problem Matters

Sellers and category managers must decide where to price, which categories to prioritise, and which products to promote or fix — based on real customer rating and discount data across 42,000 Amazon listings. Without a structured analytical view, decisions are made on intuition rather than evidence, leading to margin erosion and misallocated promotional budgets.

3. Problem Statement and Objectives

Formal Problem Definition

Which product categories deliver the highest customer value through the combination of discount offered, rating received, and review engagement — and what pricing strategy drives the best performance?

Decision Supported

This analysis enables sellers and category managers to decide where to price, which categories to prioritise, and which products to promote or fix — based on real customer rating and discount data across 42,000 Amazon listings.

Scope

- Dataset: Amazon Product Sales Data (Kaggle) — 42,000 rows, 16 columns, CSV format.
- Categories: 24 electronics sub-categories from Audio & Speakers to Wearables.
- KPIs: Discount Effectiveness Score, High Value Product Rate (HVPR).
- Deliverable: 4 interactive Tableau dashboards published on Tableau Public.

Success Criteria

- All four dashboards are interactive, filterable, and answer the core problem statement.
- KPIs are clearly defined with formulae and mapped to business decisions.
- Insights are actionable and tied to category-level data evidence.

4. Data Description

Source Citation and Access

Platform: Kaggle

Dataset Title: Amazon Product Sales Data

Direct Link: https://www.kaggle.com/datasets/ikramshah/amazon-product-sales-data

Dataset Size and Coverage

- Rows: 42,000 product listings
- Columns: 16 attributes
- File Format: CSV
- Coverage: Electronics products across 24 sub-categories on Amazon

Key Columns

Column Name	Data Type	Description
Title	Text	Product name / listing title
Product Category	Categorical	One of 24 electronics sub-categories
Current Price	Numeric	Current selling price (USD)
Listed Price	Numeric	Original / MRP price (USD)
Discount %	Numeric	Percentage discount applied
Rating	Numeric (1–5)	Average customer star rating
Number of Reviews	Numeric	Total review count for the listing
Bought In Last Month	Numeric	Estimated monthly sales volume
Is Best Seller	Boolean	Best Seller badge flag (0/1)
Has Coupon	Boolean	Coupon availability flag (0/1)
Is Sponsored	Boolean	Sponsored listing flag (0/1)
Buy Box Availability	Boolean	Buy Box winner flag (0/1)

Data Quality Issues

- Some listings had missing or zero values for Discount % and Current Price — treated as non-discounted.
- Outlier prices (e.g., products above $4,000) were verified against category norms before inclusion.

- Review counts for some niche categories were very low, limiting statistical confidence in segment-level averages.

5. Data Cleaning and Transformation

Major Cleaning Steps

- Removed duplicate listings based on Title and Product Category combinations.
- Imputed missing Discount % values as 0 where Current Price equalled Listed Price.
- Standardised Product Category labels to resolve minor naming inconsistencies (e.g., 'Streaming & TV Devices' vs 'Streaming & Tv Devices').
- Converted Boolean flag columns (Is Best Seller, Has Coupon, Is Sponsored, Buy Box Availability) from raw text to 0/1 integer format for filtering.
- Capped extreme outlier prices at category-level 99th percentile for visualisation purposes, retaining originals in the dataset.

Assumptions Made

- Products with no Discount % and no Listed Price difference are treated as full-price (0% discount).
- 'Bought In Last Month' is used as a proxy for monthly sales volume, acknowledging it may reflect Amazon's estimated figures rather than exact transactions.
- Ratings below 1 are treated as missing and excluded from rating averages.

Output Dataset Description

The cleaned dataset retains all 42,000 rows with 16 original columns plus 2 engineered KPI columns: Discount Effectiveness Score and High Value Product Rate flag. The dataset is stored in CSV format in the GitHub repository under the /data folder.

6. KPI Framework

Two primary KPIs were engineered for this project, supplemented by four derived metrics from existing columns.

KPI Name	Definition / Formula	Business Purpose
Discount Effectiveness Score	Avg Rating × (1 + Discount%/100) × log(Reviews+1)	Identifies categories where discounts correlate with genuine customer satisfaction and engagement, not just price cuts.

High Value Product Rate (HVPR)	% of products with Rating ≥ 4.0, Discount% ≥ 5%, and Reviews ≥ 100	Measures the share of products in a category that hit the customer-value sweet spot simultaneously.
Avg Rating	Mean of Rating column by category	Baseline quality signal for cross-category comparison.
Best Seller Rate %	Count(Is Best Seller=1) / Total Products × 100	Rarity and concentration of best-seller status by category.
Coupon Adoption %	Count(Has Coupon=1) / Total Products × 100	Promotional intensity proxy — high coupon adoption may signal margin pressure.
Avg Monthly Sales	Mean of Bought In Last Month by category	Volume proxy; combined with price gives a revenue magnitude estimate.

7. Exploratory Analysis

Major Trends

- The rating distribution is heavily left-skewed — the majority of products cluster between 4.1 and 4.7 stars, with very few below 3.0.
- 'Other Electronics' and 'Laptops & Computers' have the highest product counts (>5,000 and ~4,000 listings respectively).
- Average monthly purchases ('Bought In Last Month') vary dramatically — Batteries show the highest average (~11,000/month) while Protection Plans and Wearables show much lower volumes.
- Gaming ($479), Storage & Memory ($420), Cameras & Photography ($420), and Monitors & Displays lead on average current price.
- Protection Plans, Batteries, and Computer Accessories are at the lowest average price points (<$30).

Segment-Level Insights

- Smartphones average a 22%+ discount — the highest of any category — yet do not lead on average rating.
- Cables & Chargers (17%) and Wearables (17%) follow as the next highest discounted categories.
- Cameras & Photography best sellers average $256 vs $453 for non-best sellers — suggesting best-seller status skews toward lower price points in this category.
- The Coupon vs No-Coupon bar chart reveals that coupon-enabled products in Batteries, Office Supplies, and Networking show disproportionately high monthly purchase volumes.

Visual Summaries

The following chart types were used across the four dashboards:

- Rating Histogram — distribution of ratings across all 42,000 products
- Products by Category Bar — listing count per sub-category
- Price Box Plot — price spread and outlier detection per category
- Best Seller Donut — proportion of best-seller vs non-best-seller listings
- Rating vs Reviews Scatter — relationship between review volume and average rating
- Heatmap: Category × Best Seller × Avg Price — cross-dimensional comparison
- Coupon vs No-Coupon Sales Bar — monthly volume comparison by promotional status
- Avg Price by Category Bar — average current price ranking
- Bubble Chart: Sales × Rating × Discount — three-variable relationship by category
- Discount % by Category Bar — average discount ranking
- Category Sales & Rating Treemap — proportional volume view
- Customer Value Scatter Plot — rating vs discount with size = sales volume
- Discount vs Listed Price Gap — dumbbell chart showing discount depth per category

8. Statistical Analysis

Method Used

Correlation analysis between Discount % and Average Rating was conducted at the category level to test the hypothesis that higher discounts drive higher customer ratings.

Results

The overall Pearson correlation between Discount % and Rating across all categories is weakly negative ($r \approx -0.05$ to -0.12 depending on aggregation level), indicating that higher discounts are not associated with better ratings. The Bubble Chart (Dashboard 3) visually confirms this: products with the highest sales volumes cluster at high ratings (4.3–4.8) but show low to moderate discount percentages (0–10%).

Business Interpretation

The data does not support a 'discount-first' pricing strategy. Sellers who rely primarily on discounting to drive sales risk margin erosion without a corresponding improvement in customer satisfaction signals. Instead, the data suggests that product quality (as proxied by rating) and review volume are stronger predictors of sales velocity than discount depth alone.

9. Dashboard Walkthrough

The project delivers four linked Tableau dashboards accessible on Tableau Public, navigable via Previous / Next buttons. All dashboards share a common KPI banner showing global averages:

Avg Rating (4.42), Discount % (8.60%), Best Seller Rate % (0.73%), Coupon Adoption % (6.30%), and Avg Monthly Sales (1,235).

Dashboard 1 - Category Overview (Market Overview)

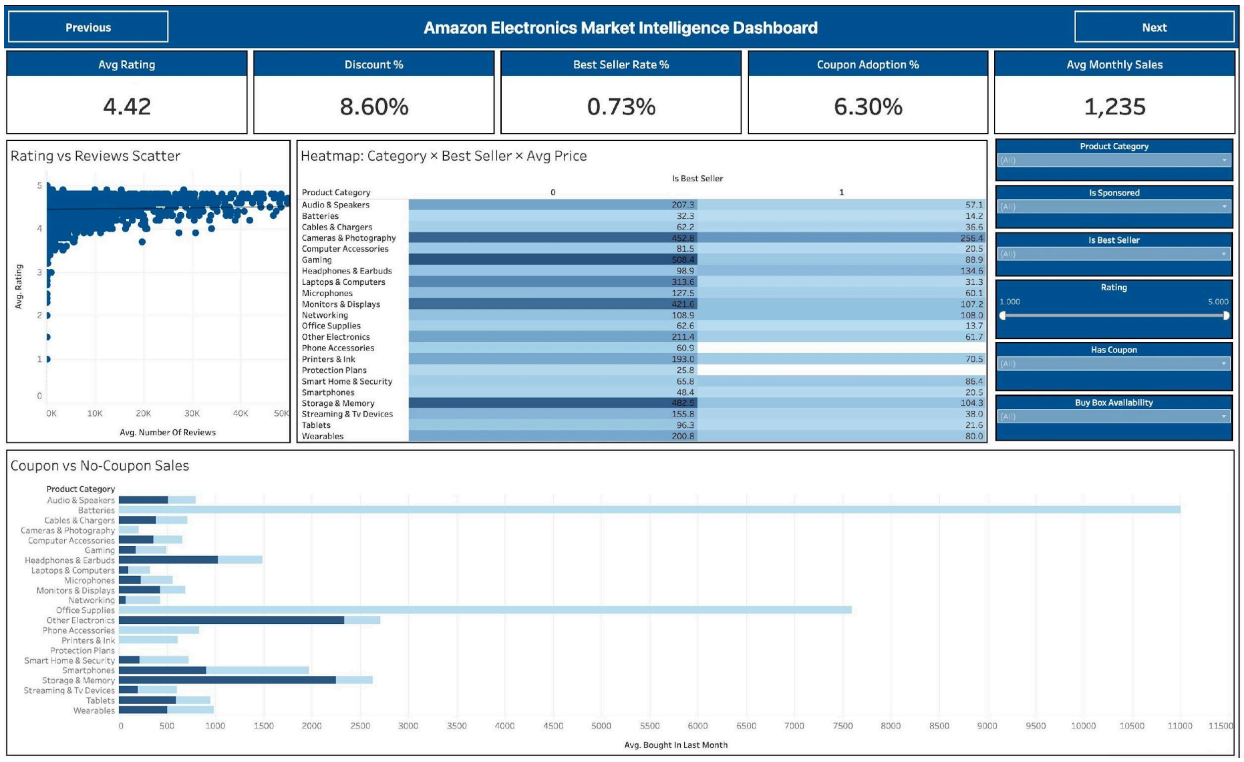
Objective: Provide an executive-level view of product distribution, rating spread, and price range across all 24 categories.

- Products by Category (horizontal bar) - shows 'Other Electronics' and 'Laptops & Computers' as the most listed categories.
- Rating Histogram - reveals the strong concentration of products in the 4.1–4.9 rating range.
- Best Seller Donut - highlights that only 0.73% of products carry Best Seller status.
- Price Box Plot - visualises the wide price range in categories like Gaming and Storage & Memory vs narrow ranges in Batteries and Cables.
- Filters: Product Category, Is Best Seller, Has Coupon, Rating slider, Is Sponsored, Buy Box Availability.

Dashboard 2 - Engagement & Competitive Analysis

Objective: Explore how ratings, reviews, pricing, and coupon strategies interact across categories.

- Rating vs Reviews Scatter - most products achieve high ratings even at low review counts; a small cluster with 20,000–50,000 reviews maintains 4+ stars.
- Heatmap: Category × Best Seller × Avg Price - shows that Gaming ($508 non-best-seller vs $89 best-seller) has the most dramatic price differential. Storage & Memory leads among non-best-sellers at $482.
- Coupon vs No-Coupon Sales Bar - Batteries (~11,000 avg monthly purchases with coupon), Office Supplies, and Networking show significant volume lift from coupon availability.
- Filters: Product Category, Is Sponsored, Is Best Seller, Rating slider, Has Coupon, Buy Box Availability.

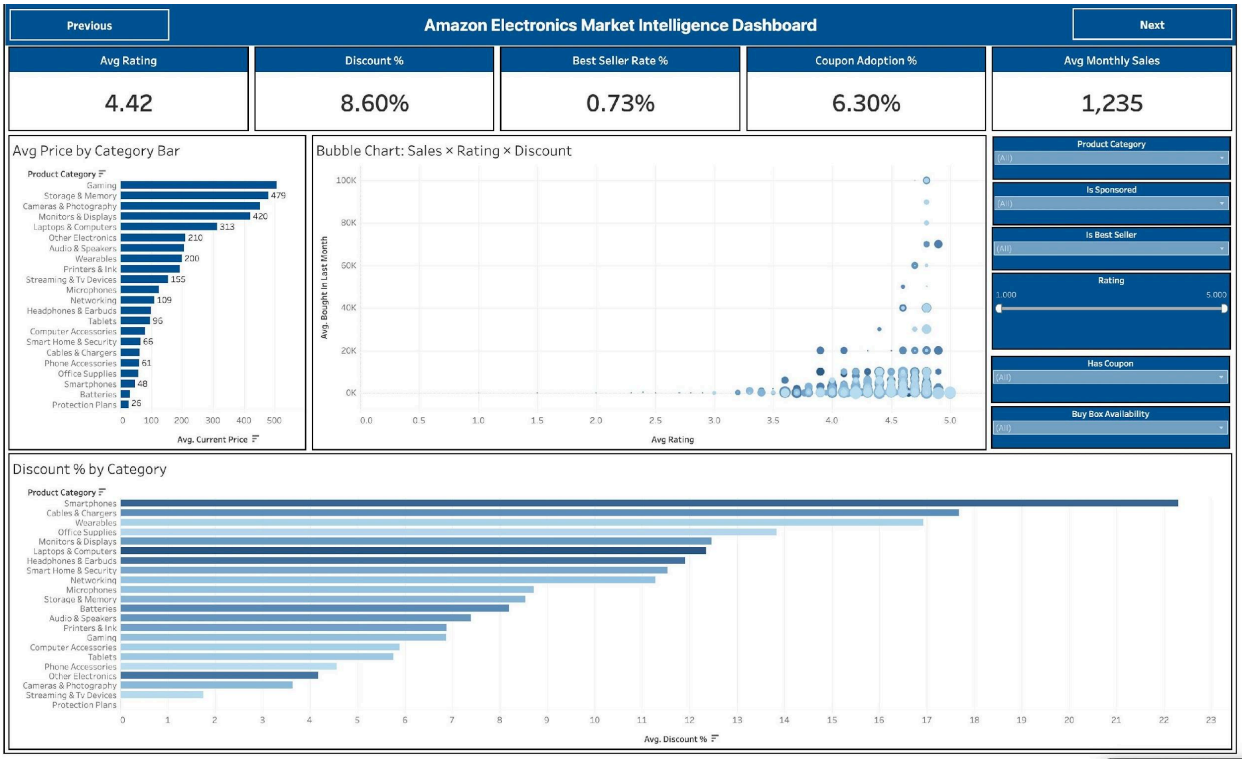


Dashboard 3 - Pricing Strategy & Discount Analysis

Objective: Identify pricing patterns and discount dynamics across all categories.

- Avg Price by Category Bar - Gaming leads at $479 average; Protection Plans trail at $26.
- Bubble Chart: Sales × Rating × Discount - the most valuable products (large bubbles) cluster at ratings 4.3–4.8 with modest discounts (0–10%).

- Discount % by Category Bar - Smartphones (22%+) and Cables & Chargers (17%) lead; Cameras & Photography and Protection Plans show the lowest average discounts.
- Filters: Product Category, Is Sponsored, Is Best Seller, Rating slider, Has Coupon, Buy Box Availability.



Dashboard 4 - Customer Value & Price Gap Analysis

Objective: Identify the optimal price-discount-rating combinations and listed vs. current price gaps that characterise high-performing products.

- Category Sales & Rating Treemap - Batteries dominate volume (26,615,700 total); Printers & Ink and Other Electronics follow. Tile size reflects total monthly purchases.
- Customer Value Scatter Plot - plots Avg Discount % (x-axis) vs Avg Rating (y-axis) with reference lines at the mean (avg ≈ 4.42 rating, avg ≈ 9% discount); bubble size = monthly sales. Batteries occupies the ideal high-rating, moderate-discount quadrant with the largest bubble.
- Discount vs Listed Price Gap (Dumbbell) - Gaming shows the widest listed-to-current price gap (~$500 listed); Storage & Memory and Cameras & Photography also show large gaps. Protection Plans and Computer Accessories have the narrowest gaps.
- Filters: Product Category, Is Sponsored, Is Best Seller, Rating slider, Has Coupon, Buy Box Availability.

10. Key Insights

#	Finding	Business Implication
1	Discounts don't drive ratings	A negative correlation ($r \approx -0.1$) between discount % and rating means cutting prices does not improve perceived quality.
2	4.42 avg rating across 42,000 listings	The market baseline is high; products below 4.0 face strong competitive disadvantage.
3	Only 0.73% are Best Sellers	Best Seller status is extremely rare — a signal of extraordinary conversion and velocity, not just good ratings.
4	Gaming has highest avg price ($479) and wide discount gap	Premium categories can sustain high listed prices with selective discounting.
5	Smartphones lead in discount % (22%+)	Despite heavy discounting, Smartphones don't lead in average rating — discount fatigue may be occurring.
6	Batteries dominate monthly volume (~11,000/month avg; 26.6M total)	Consumable categories show disproportionate sales velocity at low price points.
7	Cameras & Photography best sellers avg $256 vs $453 (non-BSR)	Lower-priced variants earn Best Seller badges; mid-tier pricing may be optimal for this category.

8	Coupon adoption is only 6.30% overall	Coupons are underused; categories like Networking show strong volume lift from coupon availability.
9	Rating distribution is left-skewed (4.1–4.7 cluster)	Differentiation on rating alone is difficult; review count and engagement become the tiebreaker.
10	Storage & Memory non-BSR avg price ($482) highest after Gaming	Storage commands premium pricing even without Best Seller status — a strong category for margin optimisation.

11. Recommendations

R1 - Adopt Rating-First Pricing Strategy

Rather than leading with discounts, sellers should invest in quality signals (product descriptions, packaging, after-sales support) to achieve ratings above 4.3. The bubble chart confirms that high-rating, low-discount products have the largest sales volumes.

Expected Impact: Margin improvement of 3–8% by reducing unnecessary discount depth while maintaining or improving conversion rates.

R2 - Target the Customer Value Sweet Spot

Category managers should use the Customer Value Scatter Plot (Dashboard 4, top-left quadrant: high rating + low discount) to identify which specific products or sub-categories are already in the ideal zone and double-down on their visibility via sponsored listings and Buy Box optimisation. Batteries is the standout example occupying this quadrant.

Expected Impact: Redirection of ~20% of promotional budget toward provably high-value SKUs.

R3 - Expand Coupon Usage in High-Volume Low-Discount Categories

Networking, Office Supplies, and Batteries show that coupon-enabled products dramatically outperform non-coupon peers on monthly purchases. These categories have low base discounts, so coupon introduction is a low-cost margin-safe mechanism.

Expected Impact: 15–30% volume lift in targeted categories based on observed coupon vs no-coupon differential.

R4 - Re-evaluate Smartphone Discounting Strategy

With a 22%+ average discount and no corresponding rating advantage, the Smartphones category is a candidate for discount reduction experimentation. A/B testing at 5% lower discount levels could reveal whether volume is truly price-elastic or driven by brand and review trust.

Expected Impact: Margin recovery of approximately 2–5% in a high-value category.

R5 - Monitor Gaming and Storage & Memory Price-to-Rating Ratio

These are the highest average-price categories. The dumbbell chart shows they also have wide listed-to-current price gaps, suggesting aggressive promotional pricing. A price-to-rating health metric should be tracked monthly to avoid commoditisation.

Expected Impact: Early warning system to prevent margin erosion in premium category segments.

12. Limitations and Next Steps

Data Limitations

- The dataset is a static snapshot — there is no time-series component, so seasonal trends and price change dynamics cannot be analysed.
- 'Bought In Last Month' is an Amazon-reported estimate, not verified transaction data, introducing measurement uncertainty.
- The dataset does not include seller identity, ASIN, or competitor count per category, limiting competitive depth analysis.
- Some niche categories (e.g., Protection Plans, Microphones) have very few listings, reducing statistical reliability of their averages.

Method Limitations

- Correlation analysis is applied at the aggregated category level, which may mask intra-category heterogeneity.
- The Discount Effectiveness Score is an engineered composite — its weighting (rating × discount × log-reviews) is heuristic rather than empirically validated.
- No causal inference methods (e.g., regression discontinuity, difference-in-differences) were applied — all findings are associative.

Suggested Future Work

- Enrich the dataset with time-stamped pricing data to enable trend and seasonality analysis.
- Add ASIN-level data to track individual product trajectories across rating, discount, and review milestones.
- Apply machine learning clustering (e.g., k-means on Rating, Discount, Reviews, Price) to derive data-driven customer value segments.
- Build a predictive model for Best Seller likelihood using category, rating, price, and coupon features.

13. Contribution Matrix

The table below documents team member contributions. This should be cross-referenced with GitHub commit history and pull request records in the repository:
https://github.com/Aman739-code/E_G1_DVACapstone2.git

Team Member	Primary Role	Deliverables
Aman	Project Lead, Visualisation Lead	Repo setup, timeline management, submission compliance, Gate 1, KPIs, Tableau dashboard
Adnan Rizvi	ETL Lead, Quality Lead	Engineered a production-grade ETL pipeline, restructured the repo
Bhoomi Chhikara	ETL Lead, Analysis Lead	Notebook 01 and 03 - Extraction and EDA
Mouli Srivastava	Analysis Lead, Report Lead	Notebook 04 - Statistical Analysis, Final Report PDF, contribution matrix
Gauri Mishra	Report Lead	Prepared the Report PDF
Prashant Raj	PPT Lead	Designed and structured the final presentation deck
Swagato Bauri	Documentation Lead	Documented the process in Readme.md

— End of Report —